OPENFOAM FOUNDATION 13

Case Management & Post-Processing Field Guide

A practical reference for the cusped leading-edge airfoil CFD case

What this guide is for

A repeatable terminal-first workflow: reopen the case, preserve results, check numerical health, calculate derived fields, sample and plot data, then work toward pressure and force coefficients. Written for OpenFOAM Foundation v13 and the case built in this project.

How to use it

Keep this guide beside the case for the next several working sessions. Do one lesson at a time. Run a command, inspect the output, write down what it means, and only then move on. Most commands below are read-only post-processing operations unless the text says they write a field or edit a dictionary.

Project path

```
~/Projects/cusped-airfoil/openfoam-case
```

Core principle

A successful solver run is a starting point, not proof of physical accuracy. Every result should be checked for numerical health, physical meaning, mesh/domain sensitivity and repeatability.

Prepared: 29 October 2026 | Based on the actual case history plus the OpenFOAM Foundation v13 User Guide.

Quick start: return to the case

Use this at the beginning of any new terminal session. Closing a terminal does not delete the saved case or results.

1. Load OpenFOAM and enter the case

```
source /opt/OpenFOAM/OpenFOAM-13/etc/bashrc
cd ~/Projects/cusped-airfoil/openfoam-case
```

2. Check saved times and current status

```
foamListTimes
printf "\nLatest saved directories:\n"
ls -d 0 [0-9]* 2>/dev/null | sort -V
printf "\nRecent solver output:\n"
tail -n 25 log.foamRun
```

The saved times currently include 5e-06, 1e-05, 1.5e-05 and 2e-05 seconds. These are snapshots from one transient run, not four separate simulations.

3. Typical post-processing commands

```
foamPostProcess -solver shockFluid -latestTime -func MachNo
foamPostProcess -solver shockFluid -latestTime -func "mag(U)"
foamPostProcess -solver shockFluid -latestTime -func vorticity
```

These calculate derived fields from saved data. They do not advance the flow solution. `-latestTime` means the most recent saved time; replace it with `-time 2e-05` when you need a specific snapshot.

4. Find the products

```
ls -lh 2e-05/Ma 2e-05/
find postProcessing -maxdepth 4 -type f -print | sort
```

Safety rule

Do not use foamCleanCase, foamListTimes -rm, or delete time directories casually. They can remove the mesh, results, or post-processing products. Back up first and inspect any cleanup script before running it.

1. Know your case

An OpenFOAM case is a directory containing configuration, mesh, and fields. Keep the distinction straight: `foamRun` writes solver results to time directories; post-processing utilities may add derived fields to time directories or write tables under `postProcessing/`. [1][2]

```
openfoam-case/
├── 0/                  initial fields and boundary conditions
├── 5e-06/, 1e-05/, ...     saved solution snapshots
├── constant/
│   ├── polyMesh/           current mesh
│   ├── physicalProperties   gas / thermophysical properties
│   └── momentumTransport    laminar model selection
├── system/
│   ├── controlDict         solver module, time controls, output
│   ├── fvSchemes           discretisation schemes
│   ├── fvSolution          equation solver controls
│   ├── blockMeshDict       structured mesh definition
│   ├── extrudeMeshDict     extrusion definition
│   └── graphUniform        line-sampling settings
├── postProcessing/         sampled data and diagnostics
├── log.foamRun             solver log
└── log.graphUniform        line-sampling log
```

Current case facts - recorded from your logs

Item	Current value / status
Solver module	shockFluid, OpenFOAM Foundation 13
Flow model	Compressible, ideal gas, Sutherland transport; laminar
Initial freestream	U = (433.9, 0, 0) m/s; T = 300 K; p field set near 1775.544 Pa
Mesh	24,000 hexahedral cells; 2D in X-Z with one thin Y layer
Mesh diagnostics	Max skewness 3.082; max non-orthogonality 87.27 degrees; 186 faces >70 degrees
Run	0 to 2e-05 s; deltaT = 1e-7 s; writes every 5e-06 s
Post-processing completed	Ma, mag(U), vorticity, global field statistics, graphUniform line file and plots

Important limitation

The latest checkMesh printed “Mesh OK”, but 87.27 degrees maximum non-orthogonality is still a serious quality warning for gradient-sensitive quantities. The domain is also much smaller than the paper’s final 80-chord-diameter far field. Treat current results as exploratory, not validated reproduction.

2. Standard session workflow and record-keeping

Start a session

```
source /opt/OpenFOAM/OpenFOAM-13/etc/bashrc
cd ~/Projects/cusped-airfoil/openfoam-case
mkdir -p notes plots scripts reports logs
```

The `mkdir -p` line is safe to repeat. Use `notes/` for session notes, `plots/` for exported figures, `scripts/` for Python/helper scripts, and `logs/` for named logs or copies of important outputs.

Save important commands and outputs

```
foamRun 2>&1 | tee log.foamRun
foamPostProcess -solver shockFluid -latestTime -func MachNo 2>&1 | tee logs/mach-latest.log
foamPostProcess -solver shockFluid -latestTime -func graphUniform 2>&1 | tee logs/graph-latest.log
```

`tee` displays output and saves a copy. For shell pipelines, `set -o pipefail` helps preserve a failing command's exit status. Use a new descriptive log filename for separate attempts so you can compare them.

```
set -o pipefail
foamPostProcess -solver shockFluid -latestTime -func MachNo 2>&1 | tee logs/mach-$(date +%Y%m%d-%H%M%S).log
```

Back up before editing

```
cp system/controlDict system/controlDict.bak-$(date +%Y%m%d-%H%M%S)
cp system/graphUniform system/graphUniform.bak-$(date +%Y%m%d-%H%M%S)
```

Use the actual dictionary name in the backup. Preserve the original STL, point data, all edited dictionaries and the solver log. A short note should record: date, objective, files changed, command run, output path, result, and next question.

A good one-line session note

```
2026-10-29 | sampled p,T,U on line z=0.052 | graphUniform | output: postProcessing/graphUniform/2e-05/line.xy |
disturbance visible near x~0.65-0.8 m; exploratory only
```

3. Solver run vs. post-processing

Operation	Purpose	Does it advance CFD time?
foamRun	Solves the flow equations and writes snapshots.	Yes
foamPostProcess ... -func MachNo	Calculates a Mach-number field from saved fields.	No
foamPostProcess ... -func "mag(U)"	Calculates speed magnitude.	No
foamPostProcess ... -func vorticity	Calculates curl of velocity.	No
foamPostProcess ... -func graphUniform	Samples a configured line from saved fields.	No
Python / Matplotlib	Reads extracted numbers and creates plots.	No

Why four time folders appeared

Your `controlDict` used `deltaT = 1e-7 s`, `endTime = 2e-5 s`, and `writeInterval = 5e-6 s` with run-time-based output. That is 200 steps total and one write every 50 steps, yielding four result snapshots (5, 10, 15 and 20 microseconds).

How to extend a finished simulation later

1. Back up `system/controlDict` and the current case before changing it.
2. Edit `startFrom` to `latestTime` and increase `endTime` beyond the latest saved time. Check `writeControl`, `writeInterval`, `deltaT` and any time-step adjustment settings.
3. Run `foamRun` again and write to a new log, e.g. `tee -a logs/extended-run.log`. Monitor Courant number, residuals, field bounds and errors.

```
grep -nE 'solver|startFrom|startTime|stopAt|endTime|deltaT|writeControl|writeInterval|adjustTimeStep|maxCo'
system/controlDict
```

Do not confuse resuming with post-processing

For another Mach field or a new line profile, keep the saved result and use foamPostProcess. Extend foamRun only when you deliberately want more simulated physical time.

4. Lesson 1 - numerical health first

Before looking at contours or reporting a coefficient, ask: did the solver finish cleanly, are variables bounded, is the time-step behaviour sensible, and is the mesh suitable? A low linear residual is not proof that the physics is correct.

Commands

```
tail -n 50 log.foamRun
grep -E "Time =|Courant Number|Solving for|ExecutionTime|FOAM FATAL|End$" log.foamRun
foamListTimes
checkMesh
```

How to read the run log

Output	What it tells you	What it does not prove
End	Solver reached configured end time without a fatal exit.	Physical convergence or validation.
Residual	How the linear system solve converged for that variable in that step.	Global or time-independent solution accuracy.
Courant number	Time-step / cell-crossing scale; high values can challenge stability or temporal accuracy.	A universal accuracy guarantee from one number.
ExecutionTime	CPU time accumulated by the run.	Physical simulated duration.
Field minima/maxima	Detects unexpected values or spikes.	Where those extrema occur, unless spatially mapped.

Current interpretation

The recorded run finished at 2e-05 s and max Courant number was about 0.485. This is evidence of a completed startup test. The simulated time is short: at about 434 m/s, the flow convects only 8.7 mm in 20 microseconds, compared with a 1 m chord. Do not describe the flow as time-converged or the airfoil drag as validated.

Mesh quality is part of post-processing

```
checkMesh -writeSurfaces -surfaceFormat vtk
find postProcessing/checkMesh -type f -print
```

The last mesh had 24,000 cells, max skewness 3.082, max non-orthogonality 87.27 degrees and 186 faces above 70 degrees. It passed the default check, but high non-orthogonality may undermine gradients, vorticity and wall quantities. A result that changes strongly with mesh refinement is not a reliable result.

5. Lesson 2 - calculate derived fields

Derived fields are calculated from the saved solution. They are not new solver runs. Run one calculation at a time, read the console message, and check the expected output path.

Mach number

```
foamPostProcess -solver shockFluid -latestTime -func MachNo
ls -lh 2e-05/Ma
```

The console should report that MachNo writes field `Ma`. This is local Mach number per cell. Its global mean is not the freestream Mach number because some cells are affected by the airfoil.

Speed magnitude

```
foamPostProcess -solver shockFluid -latestTime -func "mag(U)"
ls -lh "2e-05/mag(U)"
```

U is a vector. `mag(U)` creates the scalar speed |U| = sqrt(Ux^2 + Uy^2 + Uz^2), in m/s. Use this when direction is not the focus.

Vorticity

```
foamPostProcess -solver shockFluid -latestTime -func vorticity
ls -lh 2e-05/vorticity
```

Vorticity is curl(U), with units s^-1. Large values commonly occur where velocity gradients are steep, e.g. near a no-slip wall or shear layer. A very large peak can also be sensitive to mesh defects; locate the extremum before attributing it to physical flow.

List the tools available in this build

```
foamPostProcess -list
foamPostProcess -help
```

Foundation v13 lists field tools including MachNo, mag, vorticity, writeCellCentres, cellMin/cellMax, volAverage, forceCoeffsCompressible and graphUniform. Use the exact installed list if a name differs or a function needs more settings. [2][3]

6. Lesson 3 - field statistics and location

Minimum, maximum, mean and median help describe the spread of a field. But cell values are stored in mesh order: the first few rows of a field file are not necessarily the cells nearest to the leading edge. Global statistics do not tell you where an extremum is.

Check a field header

```
head -n 30 2e-05/Ma
head -n 30 2e-05/p
```

Simple scalar-field statistics (ASCII nonuniform List<scalar>)

```
python3 - <<'PY'
from pathlib import Path
import re, statistics
for name in ("p", "T", "Ma"):
    path = Path("2e-05") / name
    if not path.exists():
        print(name, "not found"); continue
    text = path.read_text()
    m = re.search(r"internalField\s+nonuniform\s+List<scalar>\s+(\d+)\s*\((.*?)\)\s*;", text, re.S)
    if not m:
        print(name, "field format not recognised"); continue
    n = int(m.group(1))
    v = [float(q) for q in re.findall(r"[-+]?(?:\d+\.?\d*|\.\d+)(?:[eE][-+]?\d+)?", m.group(2))]
    if len(v) != n:
        print(name, "count mismatch", n, len(v)); continue
    print(f"{name}: cells={n}, min={min(v):.6g}, max={max(v):.6g}, mean={statistics.mean(v):.6g},
median={statistics.median(v):.6g}")
PY
```

This helper is deliberately limited to ASCII scalar fields with a nonuniform internalField. If it says the format is not recognised or the counts do not match, stop and inspect the header instead of trusting partial statistics.

To locate a high/low value

```
foamPostProcess -solver shockFluid -latestTime -func writeCellCentres
ls -lh 2e-05/C*
```

The `writeCellCentres` tool writes cell-centre coordinates, which can be joined to scalar values in Python for thresholding or locating extrema. Check the actual filenames with `ls`; field filenames may vary by release/tool output.

Case-specific observation

The earlier vorticity peak was about 3.5e6 s^-1. A wall-scale velocity difference divided by a 1e-4 m length can naturally be millions per second, but this order-of-magnitude estimate does not identify the peak location or validate it. Map the peak to coordinates and test mesh sensitivity.

7. Lesson 4 - line sampling with graphUniform

A line profile turns a field into columns of values along a prescribed path. It can reveal how p, T and velocity vary with position, but it only says what happens along that line. A poorly chosen line may return only uniform freestream data.

Inspect / edit the line definition

```
cat system/graphUniform
```

The local file currently defines 200 points along x = -0.2 to 1.5 m, with fields p, T and U (Ma was requested in a configuration attempt but did not appear in the actual line.xy header).

Recommended 2D sampling plane

> **Correction to the earlier sample**
>
> The 2D mesh varies in X and Z and has a single thin Y layer with front/back empty patches. The previous line used y = -0.05 m, which is on the empty-patch plane. For the next run, sample at y = 0, where the single layer's cell centres lie. Preserve the old line file for comparison; do not delete it.

```
cp system/graphUniform system/graphUniform.bak
nano system/graphUniform
```

Use the following geometry for the near-upper-surface line. It is 3.05 mm above the maximum airfoil height at x = 0.75 m, so it is a near-wall line, not a surface sample:

```
start        (-0.2 0 0.052);
end          (1.5 0 0.052);
nPoints      200;
fields       (p T U);
axis         x;

#includeEtc "caseDicts/functions/graphs/graphUniform.cfg"
```

Run the sample and find its data

```
foamPostProcess -solver shockFluid -latestTime -func graphUniform 2>&1 | tee logs/graph-near.log
find postProcessing/graphUniform -type f -print | sort
head -n 8 postProcessing/graphUniform/2e-05/line.xy
```

Your previous line.xy header was `x p T U_x U_y U_z`. Because Ma did not appear in that actual output, calculate sampled Mach in Python from sampled U and T, or separately confirm an appropriate sampling configuration. Never infer a column is present without checking the file header. [4]

8. Plot the sampled profile in Python

Python plotting is separate from the OpenFOAM environment. If `numpy` works in one Python interpreter but not another, it usually means packages were installed into a different environment. Use a dedicated virtual environment to make it repeatable.

One-time setup (skip if the selected environment already works)

```
python3 -m venv ~/.venvs/openfoam-post
source ~/.venvs/openfoam-post/bin/activate
python -m pip install --upgrade pip
python -m pip install numpy matplotlib
python -c "import numpy, matplotlib; print(numpy.__version__, matplotlib.__version__)"
```

At a future shell session: activate this virtual environment for plotting scripts. Separately source the OpenFOAM `bashrc` to use OpenFOAM tools; these are two independent environment tasks.

Plot p, T, derived Mach and Uz from line.xy

```
cat > scripts/plot_line.py <<'PY'
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

src = Path("postProcessing/graphUniform/2e-05/line.xy")
data = np.loadtxt(src, comments="#")
if data.ndim != 2 or data.shape[1] < 6:
    raise SystemExit(f"Expected x,p,T,Ux,Uy,Uz columns; got {data.shape}")
x, p, T, Ux, Uy, Uz = data[:, :6].T
R = 8314.462618 / 28.966  # J/(kg K), from molecular weight
Cv = 719.4              # J/(kg K), case value
gamma = (Cv + R) / Cv
speed = np.sqrt(Ux**2 + Uy**2 + Uz**2)
Ma = speed / np.sqrt(gamma * R * T)

fig, axes = plt.subplots(4, 1, figsize=(9, 10), sharex=True)
series = [(p, "Pressure (Pa)", 1775.544),
        (T, "Temperature (K)", 300.0),
        (Ma, "Mach number (-)", 1.25),
        (Uz, "Vertical velocity Uz (m/s)", 0.0)]
for ax, (y, label, ref) in zip(axes, series):
    ax.plot(x, y, lw=1.5)
    ax.axhline(ref, ls="--", lw=1, label=f"reference {ref:g}")
    ax.set_ylabel(label); ax.grid(True, alpha=0.3); ax.legend()
axes[-1].set_xlabel("x (m)")
fig.suptitle("Cusped airfoil - line sample at t = 2e-5 s")
fig.tight_layout()
out = Path("plots/line_profiles.png")
out.parent.mkdir(exist_ok=True)
fig.savefig(out, dpi=180)
print("Saved", out)
PY
python scripts/plot_line.py
```

Change the input path if sampling a different time. The reference lines are the configured freestream targets, not proof of accuracy. A line profile at y=0.0 should be generated after editing graphUniform; the plot from the earlier y=-0.05 sample should be labelled exploratory.

9. Lesson 5 - pressure coefficient Cp

The paper plots pressure coefficient along the airfoil. Cp is dimensionless, so it is easier to compare pressure distributions between operating conditions than dimensional pressure alone.

```
Cp = (p - pInf) / (0.5 * rhoInf * UInf**2)
```

Here p is the local static pressure, pInf is the freestream static pressure, rhoInf is freestream density and UInf is freestream speed. The denominator is dynamic pressure qInf.

Before calculating Cp

- Confirm that the `p` field and reference pressure use the same absolute/gauge convention. The paper example describes 1775.544 Pa as a gauge pressure; do not blindly use it to calculate density until the reference-pressure convention in the OpenFOAM setup is confirmed.
- Use a consistent rhoInf and UInf from the intended freestream state. For an ideal gas, rhoInf = pAbsInf/(R*TInf) only if pAbsInf is absolute pressure.
- For a comparison with the paper, sample pressure on the airfoil wall or from a correctly extracted wall patch. Our line at z=0.052 m lies above the surface; its Cp is a near-wall-line quantity, not surface Cp.

What Cp features mean

- A pressure rise can be consistent with compression or a shock; confirm it spatially and resolve the gradient before naming it a shock.
- A pressure fall can be consistent with expansion / local acceleration.
- A sharp spike at a poor-quality cell or boundary can be numerical, so check mesh sensitivity and the location.

Physics vs. numerical evidence

Our first line plot showed pressure and velocity changes around x ~ 0.65-0.8 m. That is near the airfoil's maximum-thickness location (x = 0.75 m), but it is not enough by itself to prove a shock. The current grid and short simulated time are exploratory.

10. Lesson 6 - forces, drag and lift coefficients

A contour or line plot describes flow behaviour. An aerodynamic force calculation answers the engineering question: what net force does pressure plus viscous stress exert on the airfoil wall? OpenFOAM Foundation v13 includes `forcesCompressible` and `forceCoeffsCompressible` for dimensional-pressure compressible cases. [3]

Do not run this until reference values are confirmed

The mesh uses x as the streamwise direction and z as the lift/thickness direction; y is the thin extrusion direction. For a 0-degree case, likely coefficient directions are drag +x and lift +z. The reference area for the extruded 2D mesh must consistently use the chosen span thickness (currently 0.1 m) times chord (1 m), if that is the reporting convention adopted.

Get the Foundation v13 template

```
foamGet forceCoeffsCompressible
cat system/forceCoeffsCompressible
```

If the template is copied successfully, inspect its exact v13 syntax before editing it. Configure at minimum: wall patch `aerofoil`, freestream speed, freestream density, reference length, reference area, drag/lift directions, centre of rotation and pitch axis. Do not paste a template from another OpenFOAM distribution without checking it.

Expected concepts in a compressible coefficient configuration

```
patches (aerofoil);
magUInf 433.9;
rhoInf <confirmed absolute-state density>;
lRef 1;
Aref 0.1;
dragDir (1 0 0);
liftDir (0 0 1);
CofR (0 0 0);
pitchAxis (0 1 0);
```

The fragment shows the physical inputs to verify; use the actual keys/layout from the installed v13 template. Do not treat the placeholder density as a value to paste. An incorrect rhoInf, Aref, pressure convention or direction vector makes the coefficient misleading.

After configuration

```
foamPostProcess -solver shockFluid -latestTime -func forceCoeffsCompressible 2>&1 | tee logs/forceCoeffs.log
find postProcessing -iname "*force*" -o -iname "*coefficient*"
```

Inspect the log and output files before interpreting the values. For an unsteady run, track coefficients over time and compare statistics only after the flow has reached a suitable state and time-step/mesh checks are complete.

11. Lesson 7 - time histories and repeatable workflows

Point probes vs. line samples

Method	Question answered	Typical output
graphUniform	How does a field vary along this line at one saved time?	x/position vs field values under postProcessing/graphUniform/
probes	How does a field vary with time at a chosen point?	time-value history under postProcessing/probes/
wall forceCoeffs	How do drag/lift/moment evolve with time?	coefficient histories under postProcessing/forceCoeffsCompressible/
field statistics	How wide is the field range at one time?	summary values from a script / function output

Find the v13 probe template

```
foamGet probes
cat system/probes
```

The template typically lets you specify fields and physical probe locations. A probe records temporal behaviour at a point; a line sample records spatial behaviour along a line. Foundation v13 documents both workflows. [4]

Maintain reproducibility

- Use descriptive names for outputs, scripts and logs; never rely on memory for the command that created a plot.
- Keep the original STL, source coordinate data, dictionaries, scripts, and case notes. Save the relevant log and state which time directory was processed.
- If comparing meshes or models, create separate case copies rather than repeatedly overwriting the only baseline.

```
cd ~/Projects/cusped-airfoil
cp -a openfoam-case openfoam-case_baseline-copy
```

Cleanup warning

foamCleanCase resets generated workflow files and can remove constant/polyMesh, result times and postProcessing. foamListTimes -rm deletes time results. Never run cleanup commands as a casual “reset”; first make a copy and check the installed utility help.

12. Validation checklist and next 10 sessions

Before trusting any result

- Numerical: solver reaches intended time; inspect logs, Co, residuals and field bounds.
- Physical: check freestream p/T/U/Mach against the target away from the airfoil.
- Spatial: map extrema to coordinates; use contours/line data to explain where changes occur.
- Mesh: resolve the very high non-orthogonality; compare a refined mesh and verify wall resolution.
- Domain: enlarge the current box toward the paper's far-field extent and test boundary sensitivity.
- Time: extend the run; monitor pressure / forces versus time and compare time-step sensitivity.
- Metrics: use surface pressure for Cp and the correct rhoInf, UInf, Aref and directions for force coefficients.
- Validation: compare to the paper only after setup and physical assumptions are consistent.

A 10-session learning plan

Session	Focus	Deliverable
1	Reopen case, logs, saved times	Status note and clean log
2	Derived fields: Ma, mag(U), vorticity	Verified fields and statistics
3	Global stats and locating extrema	Coordinate/value table
4	graphUniform line sampling	New sample at y=0 with data header checked
5	Python plots & interpretation	Pressure/T/Mach/Uz plots
6	Wall pressure / surface Cp	Cp vs x with reference state documented
7	Forces / coefficient template	Validated force configuration and log
8	Probe/force histories over time	Time series, identify transient behaviour
9	Mesh, time-step, domain sensitivity	Comparison table / decisions
10	Reproducible case & reporting	README, scripts, logs, assumptions and limitations

Session note template

```
Date/time:
Question / goal:
Case + time directory:
Files changed:
Commands run:
Outputs created:
Observed result (include units):
Physical interpretation:
Uncertainty / checks still needed:
Next action:
```

13. Troubleshooting quick reference

Symptom	Check this first	Typical action
foamPostProcess says field not found	Does the field exist at the selected time? `ls 2e-05/`	Choose a time that contains the field; run the appropriate derived-field function if needed.
graphUniform output lacks a requested column	Read the header: `head -n 8 .../line.xy`	Trust the header. Confirm the field exists and is loaded; derive Mach from sampled U,T if necessary.
line file is uniform	Where is the line relative to body and disturbed region?	Move the line deliberately; check axes and y=0 plane for this 2D mesh.
Python cannot import numpy	Which Python is running? `which python; python -m pip show numpy`	Activate the virtual environment used to install packages.
solver finishes but field values look strange	Log, units, pressure convention, boundary condition, mesh diagnostics	Do not interpret extrema before checking these items.
Mesh says OK but gradients are spiky	Non-orthogonality, skewness, location of gradient peak	Inspect/repair mesh; test refinement and sensitivity.
Need more simulated time	Current startFrom/endTime in controlDict	Back up; set startFrom latestTime and increase endTime, then run a new logged extension.
Output directories disappeared	Cleanup scripts or `foamListTimes -rm`	Restore from backup; add backups before future runs.

Useful safe inspection commands

```
foamPostProcess -list
foamPostProcess -help
foamDictionary system/controlDict
foamDictionary system/fvSchemes
foamDictionary system/fvSolution
foamListTimes
checkMesh
tail -n 50 log.foamRun
find postProcessing -type f -print | sort
```

If a command fails

Do not repeatedly try guessed function names or overwrite dictionaries. Capture the exact error (`tee` a log), check the local template with `foamGet`, read its contents, and use the OpenFOAM Foundation v13 documentation for that utility.

14. References and case-specific cautions

Documentation used for standard workflow and command semantics. The installed Foundation v13 build and its own tutorial files should take precedence over examples for OpenCFD/OpenFOAM.com or other versions.

[1] OpenFOAM Foundation v13 User Guide - Case management tools
https://doc.cfd.direct/openfoam/user-guide-v13/case-management

[2] OpenFOAM Foundation v13 User Guide - Post-processing CLI
https://doc.cfd.direct/openfoam/user-guide-v13/post-processing-cli

[3] OpenFOAM Foundation v13 User Guide - Post-processing functionality
https://doc.cfd.direct/openfoam/user-guide-v13/post-processing-functionality

[4] OpenFOAM Foundation v13 User Guide - Sampling and monitoring data
https://doc.cfd.direct/openfoam/user-guide-v13/graphs-monitoring

[5] OpenFOAM Foundation v13 API - MachNo function object
https://cpp.openfoam.org/v13/classFoam_1_1functionObjects_1_1MachNo.html

[6] OpenFOAM Foundation v13 User Guide - Main index
https://doc.cfd.direct/openfoam/user-guide-v13/index

[7] Akram, S. & Hasan, N. (2024), "The dynamics of supersonic flow past a new cusped leading edge airfoil," Progress in Computational Fluid Dynamics, 24(1), 1-28. DOI: 10.1504/PCFD.2022.10052925
https://doi.org/10.1504/PCFD.2022.10052925

Important details that still need to be resolved

- Pressure convention: confirm whether the 1775.544 Pa in the paper is gauge pressure and how it maps to the OpenFOAM thermodynamic pressure / absolute pressure used to calculate density.
- Far-field: the paper's final computational domain is much larger (80 chord diameters) than the current box. Perform domain sensitivity before comparing aerodynamic values.
- Mesh: max non-orthogonality is about 87.27 degrees even though the final default `checkMesh` printed Mesh OK. Do not assume the gradient and wall quantities are insensitive to this.
- Line sampling: previous line used y = -0.05 m on an empty patch plane. Re-sample at y = 0 for the 2D mid-plane and label the old data as exploratory.
- Duration: 20 microseconds is an initial startup interval, not evidence that unsteady force histories have reached repeatable statistics.
- Comparison: the paper used ANSYS Fluent and its own structured O-grid / pressure far-field conditions. The current case uses an adapted Foundation v13 approach; document differences rather than claim exact reproduction.

The standard research habit

Record what you changed, why, what the output showed, and what remains uncertain. Do not hide a warning; use it to decide the next test. A documented limitation is better than an unsupported claim.